THE HANDICAP ARENA

A Sanctioned Competition Circuit for Local AI · Architectural Specification v1.0

Logline: *Every leaderboard on the internet ranks a 4-billion-parameter model running on a six-year-old graphics card against frontier models and tells you it came 417th.* ***It didn't come 417th. It was competing in a different sport.***

8 GB	4 of 11	0 Errors	~19×	720 Lines
Reference GPU (RTX 3060 Ti)	Entrants Scratched at Weigh-in	Thrown Across 4 False Boards	Biathlon Spread from Latency	arena.py (Zero External Libs)

1. THE CRISIS: INVISIBLE BY CONSTRUCTION & EVALUATION CONFOUNDS

Traditional leaderboards rank models globally on academic benchmarks by parameter counts written in filenames. A quantised 4B model running on an 8-gigabyte consumer GPU—the most ubiquitous configuration in the world—is placed in the hundreds against cloud APIs that cost money per token and cannot run locally at all. The question local practitioners ask daily is not *'what is the best model in the world?'* but **'what is the best model that fits my hardware, for my task?'** Nobody answers that reproducibly.

Furthermore, evaluation suites frequently measure plumbing rather than intelligence. During the Arena's initial development, the harness generated **three clean, confident, entirely wrong leaderboards in a row**. Every failure mode was silent, and every one produced a legitimate-looking table without throwing an error.

#	The Confound	What It Silently Faked	Harness Failure Mode
1	**Stale server config**	Two models 'scored' 0.00 that never ran at all	Pointed at deleted model files; dead endpoints returned 404s recorded as zero.
2	**Reasoning field trap**	A strong model scored 0.00 despite solving tasks	Harness inspected empty answer field; response was in reasoning_content.
3	**Grammar file locks**	Four models scored 0.00 despite high capability	Serving schema locked output to another schema; models physically couldn't answer.
4	**Alias substitution**	Undetectable downstream model corruption	Endpoint labeled granite-8b silently served a Qwen model.

The 5-Measurement Diagnostic Proof: The same 2-billion-parameter model was measured five times during development with zero weight modifications: **0.00 -> 0.33 -> 0.67 -> 0.33 -> 0.67**. Every movement came from configuration flags (parsing the wrong field, token budget truncation, and context window limits). **The harness is the measurement.** A benchmark that does not own its server flags systematically penalizes reasoning models, because fixed token budgets cap how much a capable model is allowed to think.

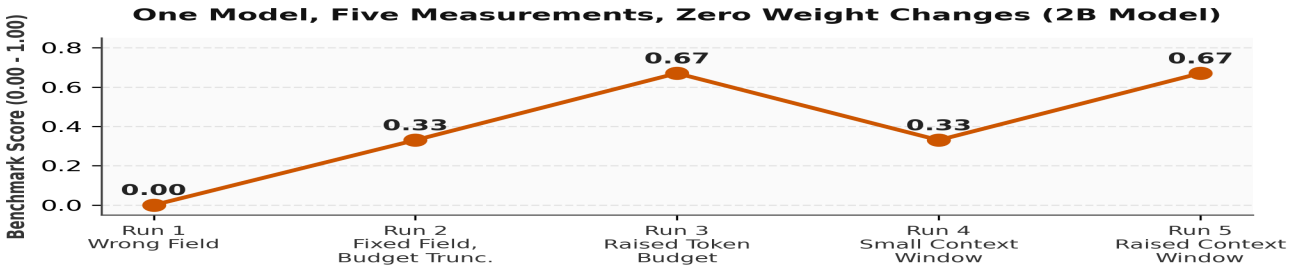


Figure 1: Benchmark score progression of an identical 2B model across five harness configurations with zero weight changes.

2. THE INSTITUTION: WEIGHED DIVISIONS & DEMONSTRATED BELTS

A local model's defining attribute is not declared parameters, but what it occupies in video memory while executing. The Arena weighs models on a digital scale: baseline-subtracted peak VRAM sampled during execution. Declared classes are guesses; weighed classes are facts. A 0.21 GB file on disk measures **0.42 GB** when served, while a 1.25 GB file measures **1.60 GB** due to KV cache allocations and runtime buffers.

Division	Measured Footprint	Cord	Operational Significance
Featherweight	≤ 4 GB	Teal	Ultra-compact models, edge devices, sub-half-gigabyte entries.
Lightweight	≤ 8 GB	Amber	**The Reference Class:** standard consumer hardware (RTX 3060 Ti).
Middleweight	≤ 16 GB	Violet	Mid-tier enthusiast cards and prosumer workstations.
Heavyweight	> 16 GB	Crimson	High-VRAM single workstation configurations.
Open / Offload	CPU + System RAM	Slate	Offloaded hybrid execution where memory bandwidth governs throughput.

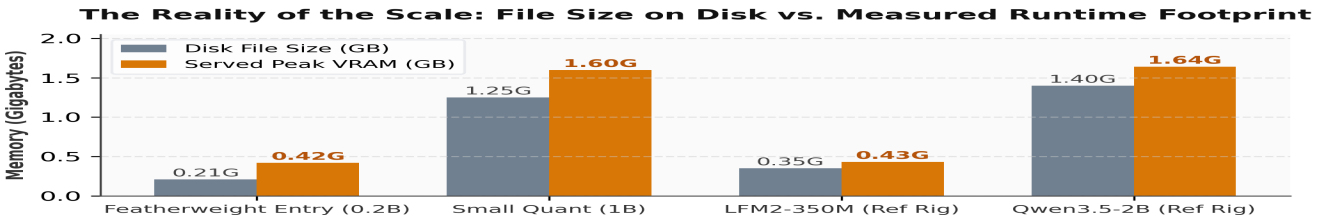


Figure 2: Physical VRAM footprint measured during inference vs declared file size on disk.

> **The Mandatory Weigh-In:** Before any bout, every entrant must prove it can successfully serve on the host hardware. A model that fails to load is **SCRATCHED — never scored**. On the first live card, 4 of 11 entrants were scratched due to missing files or unreachable endpoints. Scoring an unservable model with a 0.00 isn't a technicality; it is publishing a lie.

Rank by Demonstration: The Martial Arts Belt Ladder

Rank in the Arena is earned strictly through verified bouts, never by panel award or corporate prestige. Rankings decay automatically after 180 days without fighting to prevent legacy models from resting on stale reputation.

Belt Grade	Qualification Criteria	Dan & Defense Mechanics
White	Weighed in — successfully proved serving capability at the weigh-in scale.	Entry level.
Yellow	Won at least 1 round in a sanctioned bout.	Basic competency.
Orange	Won 2 of 3 rounds in a sanctioned bout.	Demonstrated consistency.
Green	Swept an entire 3-round bout (3-0).	Division contender.
Blue	Swept two consecutive bouts.	Proven mastery.
Purple	Swept three consecutive bouts.	**Mandatory challenge one division up.**
Brown	Ranked in the top three on a division card.	Title challenger.
Black	Division Champion.	Crown holder.
1st–10th Dan	Consecutive successful title defenses.	**Dan grade = defense count.** Cannot be gifted.
Red Belt	Held a sanctioned belt in three distinct weight divisions.	**The Summit:** Proves where capability breaks with size.

3. THE COMPETITIVE CIRCUIT: DETERMINISTIC GAMES & REAL-WORLD TRADEOFFS

Every Arena event uses an objective, ground-truthed scorer or programmatic execution verifier. **LLM-as-a-judge is strictly prohibited**—allowing a language model to grade another model is a family member marking the exam. Code is run against test suites in isolated sandboxes; financial actions are scored against realized market returns; diagrams are checked by structural similarity.

The Biathlon: Modeling Deployment Reality

Accuracy without throughput is commercially unusable. Real biathlon ranks on total time including penalty loops. That is exactly the local-inference trade-off:

Total Latency = (latency × trials) + (misses × k × latency)

The constant **k** represents the engineering penalty: how much an incorrect prediction costs relative to inference time in your application. On the Featherweight card, one model achieved 2× higher raw accuracy than its competitor, but lost the Biathlon by **~19×** because its inference latency was ~16× slower (≈2.7 s vs ≈52 s). Both numbers are true; the event reveals the real deployment trade.

Event Format	Execution Mechanics	Deployment Problem Solved
Single Bout	3 rounds, one model, one belt grade awarded.	Baseline operational competence.
Cage Match	2 models, blind, deterministic scoring. Rebuttal variant allows revisions.	Head-to-head showdown under identical constraints.
Biathlon	Accuracy × throughput; misses incur scaled penalty loops.	**The commercial deployment trade.**
Tag Team / Relay	N models; context baton passed at typed handoffs. Team weighed at sum of VRAM.	**Can chaining beat a single bigger model?**
Marathon	Long-context endurance; output evaluated as an attention decay curve.	Discovers exactly where attention breaks under load.
Decathlon	10 heterogeneous disciplines across agency, speed, code, and vision.	Comprehensive architectural profile.

Tag-Team Relays & Prediction Market Empirical Data

Production Architecture as Sport: Modern local deployments use tag-team chaining (small routers passing context to larger reasoners). In the Arena, teams compete at the **sum of their footprints** (e.g., a 350M + 2B team totals ~2 GB and fights in Featherweight; a 2B + 9B team measures ~11 GB and enters Middleweight). You cannot field three giants and call yourself a featherweight.

Empirical Finding in Market Models: Across 230 resolved prediction market tests, no local model beat the closing market price. However, a narrow **9B parameter model beat a 35B model**. Capability did not scale with raw parameter volume, demonstrating that task-specific fitness beats uncalibrated brute force.

> **Four Inviolable Rules of the Arena:**
> 1. **Deterministic Scoring:** Every round evaluated by programmatic execution or ground truth.
> 2. **Mandatory Null Baselines:** Every event includes a trivial script or random policy to expose false wins.
> 3. **Dual-Track Cost Scoring:** Every leaderboard publishes absolute score AND score per gigabyte of VRAM.
> 4. **Pre-Registration:** Scorers, datasets, and decision rules are frozen before execution to prevent post-hoc storytelling.

4. THE HARNESS ARCHITECTURE & CRYPTOGRAPHIC VERIFICATION

arena.py (~720 Lines, Zero Dependencies): Written entirely in Python standard library, the harness controls the complete execution lifecycle. It launches its own isolated server instance with controlled flags, measures baseline and peak VRAM, runs generated code against unit tests in sandboxed subprocesses, and cleans up GPU allocations upon exit. It contains 11 self-test assertions proving that scorers reject invalid answers. Pointed at legacy bundles lacking round win counts, it refuses to grade: *'CANNOT DERIVE A BELT: Reporting 0 stripes here would be a confident lie.'*

Cryptographic Certification: Eliminating Cheating by Construction

Leaderboards relying on user-submitted telemetry are inevitably gamed by quant manipulation or routed frontier APIs. The Arena operates two tiers: **Open Tier** (labeled UNCERTIFIED, screened via GGUF hashes, throughput ceilings, and inter-token jitter) and **Certified Tier** (re-running the exact binary hash on an independent reference rig). **Cheating is not caught—it is made impossible, because we re-ran the model ourselves.**

Entrant Model	Measured VRAM	Belt Grade	Stripes	Reference Rig Performance (RTX 3060 Ti 8GB)
LFM2-350M	0.43 GB	Yellow	3/9	Swept tool-calling 3/3 against larger entrants on consumer hardware.
Qwen3.5-2B	1.64 GB	Orange	6/9	High reasoning capacity; answered every round out of its reasoning field.

5. OPERATIONAL ROADMAP & CONCLUSION

#	Phase	Core Deliverable & Stop Condition	Cost Profile
0	**Market Logger**	Daily snapshot of candidate markets; time-gated foundational data.	Free (Time-gated)
1	**Pre-Register Scoreboard**	Freeze evaluation rules on forward outcomes to prevent gaming.	Free
2	**Selection Inventory**	Test strategy against null baselines (stop if random policy wins).	Free
3	**Training Toolchain**	Install fine-tuning stack (stop if model exceeds VRAM ceiling).	Hardware Bound
4	**Test Adapter**	Train, quantise, convert, and verify narrow adapter beats base model.	Moderate Compute
5	**Sanctioned League**	Launch public bouts, cards, belts, and remote certified challenges.	Ongoing

> **The Closing Architectural Thesis:**
> A local-model benchmark that doesn't publish its token budget, context window, server flags, and model hashes isn't measuring models. It's measuring plumbing — and it doesn't know it.